Program : **Diploma in Printing Technology**	
Course Code : **6102B**	Course Title: **Advancement in Packaging**
Semester : **6**	Credits: **4**
Course Category: **Program Elective**	
Periods per week: **4 (L:3, T:1, P:0)**	Periods per semester: **60**

Course Objectives:

- To provide the exposure to the essentials of advanced technologies arises in packaging.
- To guide the student to the basic concepts of the packaging trends and innovations and to learn how it influences the real printing and packaging world.
- To understand about the influence of smart concept in packaging process.
- To develop idea on exciting possibilities of various advancements in packaging.

Course Prerequisites: Nil

Course Outcomes

On completion of the course, the student will be able to:

COn	Description	Duration (hours)	Cognitive Level
CO1	Explain the smart & intelligent packaging concept.	15	Understanding
CO2	Describe the basics of various advanced packages.	14	Understanding
CO3	Explain the basics aspects of latest trends in packaging.	15	Understanding
CO4	Contrast the recent advancements, legislative issue and safety of packaging.	14	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Explain about the Smart & Intelligent packaging concept.**		
M1.01	Introduction to advancement in packaging.	3	Understanding
M1.02	Smart and intelligence packaging features.	4	Understanding
M1.03	Indicators /sensors/scavengers/emitters in packaging.	4	Understanding
M1.04	Automatic identification system.	4	Understanding
Contents: Smart and Intelligence Packaging-Introduction, definition and importance. Indicators: Integrity indicators, Time-Temperature indicators, Freshness indicators. and Sensors: Oxygen sensors, Gas sensors, Bio sensors. Scavengers & Emitters: Oxygen scavengers, Ethylene scavengers, Ethanol emitters; Carbon Dioxide scavengers/emitters, Enzymatically active packages and other technologies. Moisture control, Anti-microbial packaging, Freshness and Temperature control. Modified Atmospheric Packaging (MAP). Automatic identification systems: Barcodes; Card Technologies-Smart cards, Magnetic cards, Optical cards; Radio Frequency Identification authentication and tracking technologies for packaging; Laser surface authentication; Hologram; Security features; Security inks; Thermo-chromic labels.			
CO2	**Describe the basics of various advanced packages.**		
M2.01	Edible Packaging.	1	Understanding
M2.02	Anti-microbial Packaging.	2	Understanding
M2.03	Water soluble packaging	1	Understanding
M2.04	Self Cooling and Self heating Packaging.	2	Understanding
M2.05	Flavour and Odour Absorber.	1	Understanding
M2.06	Micro Packaging.	2	Understanding
M2.07	Skin Pack.	2	Understanding
M2.08	reCup, Compostable Coffee Capsule.	1	Understanding

<table>
<tr><td>M2.09</td><td>Bees Wrap</td><td>1</td><td>Understanding</td></tr>
<tr><td>M2.10</td><td>Direct Thermal Film Label.</td><td>1</td><td>Understanding</td></tr>
<tr><td></td><td>Series Test I</td><td>1</td><td></td></tr>
<tr><td colspan="4">Contents:
Features, advantages and packaging principles of: Edible Packaging, Anti-microbial Packaging, Water soluble packaging, Self Cooling/Self Chilling Cans and Self heating Packaging, Flavour and Odour Absorber, Micro Packaging, Skin Pack, reCup, Compostable Coffee Capsule, Bees Wrap, Direct Thermal Film Label.</td></tr>
<tr><td>CO3</td><td colspan="3">Explain the basics aspects of latest trends in packaging.</td></tr>
<tr><td>M3.01</td><td>Printed Electronics in Packaging.</td><td>3</td><td>Understanding</td></tr>
<tr><td>M3.02</td><td>Zero Waste Packaging system.</td><td>2</td><td>Understanding</td></tr>
<tr><td>M3.03</td><td>Insignia Packaging Technologies.</td><td>3</td><td>Understanding</td></tr>
<tr><td>M3.04</td><td>Atmosphere Packaging system.</td><td>2</td><td>Understanding</td></tr>
<tr><td>M3.05</td><td>3D printing in packaging.</td><td>3</td><td>Understanding</td></tr>
<tr><td>M3.06</td><td>Digitalization in packaging.</td><td>2</td><td>Understanding</td></tr>
<tr><td colspan="4">Contents:
Printed Electronics in Packaging, Features advantages and packaging systems of: Zero Waste Packaging, Insignia Technologies, Atmosphere Packaging; 3D printing in packaging, Digitalization in packaging.</td></tr>
<tr><td>CO4</td><td colspan="3">Describe the Automation in packaging systems, legislative issue and safety of packaging.</td></tr>
<tr><td>M4.01</td><td>Non-destructive inspection methods.</td><td>2</td><td>Understanding</td></tr>
<tr><td>M4.02</td><td>Printing techniques.</td><td>3</td><td>Understanding</td></tr>
<tr><td>M4.03</td><td>Application of robotics and machineries.</td><td>2</td><td>Understanding</td></tr>
<tr><td>M4.04</td><td>Automation architecture and Software systems.</td><td>2</td><td>Understanding</td></tr>
<tr><td>M4.05</td><td>Legislative issues.</td><td>2</td><td>Understanding</td></tr>
<tr><td>M4.06</td><td>General safety and Labeling.</td><td>1</td><td>Understanding</td></tr>
<tr><td>M4.07</td><td>New forms of food packaging produced using nanotechnology.</td><td>2</td><td>Understanding</td></tr>
<tr><td></td><td>Series Test II</td><td>1</td><td></td></tr>
<tr><td colspan="4">Contents:
Recent advances such as non-destructive inspection methods, printing techniques, application of robotics and machineries, automation architecture and software systems. Legislative issue, Legislation relevant to smart packaging, authorization procedure, General safety, Labelling, New forms of food packaging produced using nanotechnology.</td></tr>
</table>

Text / Reference:

T/R	Book Title/Author
T1	IIP- **Packaging technology**- Vol – I, II, III- IIP India. .
R1	Frank Paine- **Packaging design and performance**.
R2	Jhonn Briston- **Advance in plastic packaging technology**.
R3	Kit L. Yam-**Encyclopedia of Packaging Technology**- A John Wiley & Sons, Inc., Publication.

Online resources:

Sl.No	Website Link
1	https://foodpackaging.foodtechconferences.org/
2	https://www.smithcorona.com/blog/packaging-innovations
3	https://www.packagingtechtoday.com/
4	https://pakfactory.com/blog/future-of-packaging-technology
5	https://www.industrialpackaging.com/